COHOMOLOGY OF A COMPLEX OF MODULES AGREES WITH THAT OF ITS UNDERLYING COMPLEX OF ABELIAN SHEAVES, UNBOUNDEDLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. The phrase 'see discussion in Section ...' inside it is a pointer to a section of the source tree and not an omission; the closing sentence 'We omit the detailed verification that this construction is indeed an inverse.' is a routine tail. Both are kept verbatim inside the counted proof span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Unbounded derived category of modules over a ringed site and derived sections and pushforward. Same chapter, the functors RGamma and Rf_* on the unbounded derived category and their computation from K-injective representatives. Supplies the maps whose being isomorphisms is asserted.
- K-flat resolutions with flat terms and cofinality of such resolutions. Same chapter, the existence of K-flat resolutions with flat terms and the cofinality of the quasi-isomorphisms from bounded above complexes of flat modules, together with the companion isomorphism between the cohomology of a module and of its underlying abelian sheaf at a point. Supplies the construction of the inverse map in the proof.

Source reference. Original lemma 0D6F: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4329–4329. Navigation pointer invoked at proof line 4485; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0D6I: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4434–4434. Navigation pointer invoked at proof line 4486; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03EW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4518–4518. Navigation pointer invoked at proof line 4518; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05T7: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4519–4519. Navigation pointer invoked at proof line 4519; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 06YL: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2961–2961. Navigation pointer invoked at proof line 4520; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YT: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3233–3233. Navigation pointer invoked at proof line 4540; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Navigation pointer invoked at proof line 4541; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O}_\mathcal{C})$ be a ringed site. Let K be an object of $D(\mathcal{O}_\mathcal{C})$ and denote K_{ab} its image in $D(\underline{\mathbf{Z}}_\mathcal{C})$.*

1. *There is a canonical map $R\Gamma(\mathcal{C}, K) \to R\Gamma(\mathcal{C}, K_{ab})$ which is an isomorphism in $D(Ab)$.*
2. *For any $U \in \mathcal{C}$ there is a canonical map $R\Gamma(U, K) \to R\Gamma(U, K_{ab})$ which is an isomorphism in $D(Ab)$.*
3. *Let $f : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed sites. There is a canonical map $Rf_*K \to Rf_*(K_{ab})$ which is an isomorphism in $D(\underline{\mathbf{Z}}_\mathcal{D})$.*

Proof. The map is constructed as follows. Choose a K-injective complex $\mathcal{I}^\bullet$ representing K. Choose a quasi-isomorpism $\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ where $\mathcal{J}^\bullet$ is a K-injective complex of abelian groups. Then the map in (1) is given by $\Gamma(\mathcal{C}, \mathcal{I}^\bullet) \to \Gamma(\mathcal{C}, \mathcal{J}^\bullet)$ (2) is given by $\Gamma(U, \mathcal{I}^\bullet) \to \Gamma(U, \mathcal{J}^\bullet)$ and the map in (3) is given by $f_*\mathcal{I}^\bullet \to f_*\mathcal{J}^\bullet$. To show that these maps are isomorphisms, it suffices to prove they induce isomorphisms on cohomology groups and cohomology sheaves. By Lemmas 0D6F and 0D6I it suffices to show that the map

$$H^0(\mathcal{C}, K) \longrightarrow H^0(\mathcal{C}, K_{ab})$$

is an isomorphism. Observe that

$$H^0(\mathcal{C}, K) = \mathrm{Hom}_{D(\mathcal{O}_\mathcal{C})}(\mathcal{O}_\mathcal{C}, K)$$

and similarly for the other group. Choose any complex $\mathcal{K}^\bullet$ of $\mathcal{O}_\mathcal{C}$-modules representing K. By construction of the derived category as a localization we have

$$\mathrm{Hom}_{D(\mathcal{O}_\mathcal{C})}(\mathcal{O}_\mathcal{C}, K) = \mathrm{colim}_{s:\mathcal{F}^\bullet \to \mathcal{O}_\mathcal{C}} \mathrm{Hom}_{K(\mathcal{O}_\mathcal{C})}(\mathcal{F}^\bullet, \mathcal{K}^\bullet)$$

where the colimit is over quasi-isomorphisms s of complexes of $\mathcal{O}_\mathcal{C}$-modules. Similarly, we have

$$\mathrm{Hom}_{D(\underline{\mathbf{Z}}_\mathcal{C})}(\underline{\mathbf{Z}}_\mathcal{C}, K) = \mathrm{colim}_{s:\mathcal{G}^\bullet \to \underline{\mathbf{Z}}_\mathcal{C}} \mathrm{Hom}_{K(\underline{\mathbf{Z}}_\mathcal{C})}(\mathcal{G}^\bullet, \mathcal{K}^\bullet)$$

Next, we observe that the quasi-isomorphisms $s : \mathcal{G}^\bullet \to \underline{\mathbf{Z}}_\mathcal{C}$ with $\mathcal{G}^\bullet$ bounded above complex of flat $\underline{\mathbf{Z}}_\mathcal{C}$-modules is cofinal in the system. (This follows from Modules on Sites, Lemma 03EW and Derived Categories, Lemma 05T7; see discussion in Section 06YL.) Hence we can construct an inverse to the map $H^0(\mathcal{C}, K) \longrightarrow H^0(\mathcal{C}, K_{ab})$ by representing an element $\xi \in H^0(\mathcal{C}, K_{ab})$ by a pair

$$(s : \mathcal{G}^\bullet \to \underline{\mathbf{Z}}_\mathcal{C}, a : \mathcal{G}^\bullet \to \mathcal{K}^\bullet)$$

with $\mathcal{G}^\bullet$ a bounded above complex of flat $\underline{\mathbf{Z}}_\mathcal{C}$-modules and sending this to

$$(\mathcal{G}^\bullet \otimes_{\underline{\mathbf{Z}}_\mathcal{C}} \mathcal{O}_\mathcal{C} \to \mathcal{O}_\mathcal{C}, \mathcal{G}^\bullet \otimes_{\underline{\mathbf{Z}}_\mathcal{C}} \mathcal{O}_\mathcal{C} \to \mathcal{K}^\bullet)$$

The only thing to note here is that the first arrow is a quasi-isomorphism by Lemmas 06YT and 06YQ. We omit the detailed verification that this construction is indeed an inverse. □